GSoC 2026 Project Proposal

Experimentation Engine For Studying Trust Calibration

Organization: ISSR

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1 Introduction and Contributor Information

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2 Abstract

This project will build a modular web-based experimentation platform for studying trust calibration in AI-assisted decision systems. It will allow researchers to manipulate signaling cues in AI-assistants and then measure varying behavioral outcomes of specific test subjects.

The final deliverable will be an open-source prototype with participant and condition assignment, one structured decision task, fine-grained logging, JSON/CSV export, a reproducible Python analysis notebook, and clear documentation for reuse. It will be instrumental in allowing researchers to quickly, effectively, and completely gather all necessary information required in order to study the impact AI assistants cues have on behavioral outcomes.

3 Technical Details

3.1 Project Motivation

Many trust-in-AI studies rely heavily on self-report scales. While useful, self-report alone does not fully capture whether users actually rely on or override AI recommendations in decision-making settings. This project focuses instead on behavioral trust metrics through observable actions such as accepting or overriding recommendations, along with response latency and other factors found through high-fidelity logging.

The broader motivation is to provide a reusable research platform that helps distinguish trust induced by interface presentation from trust earned through actual system capability. By building modular infrastructure for controlled cue manipulation and structured logging, this project can support more rigorous and reproducible human to AI trust studies.

3.2 Proposed System Design

The system will be implemented as a modular web application using Next.js and TypeScript. The architecture will separate the experiment into the following layers:

- **Participant/session handling:** participant ID assignment and persistent condition assignment.
- **Condition management:** structured definitions for cue manipulations such as assistant name, tone, and confidence framing.
- **Task presentation:** a web interface for presenting the experimental decision task.
- **Instrumentation/logging:** collection of decision events, timestamps, and latency.
- **Export layer:** generation of JSON and CSV outputs.
- **Analysis layer:** a reproducible Python notebook for summary analysis.

This modular design keeps the initial implementation scoped while making it straightforward to add new cues, task types, or richer instrumentation later.

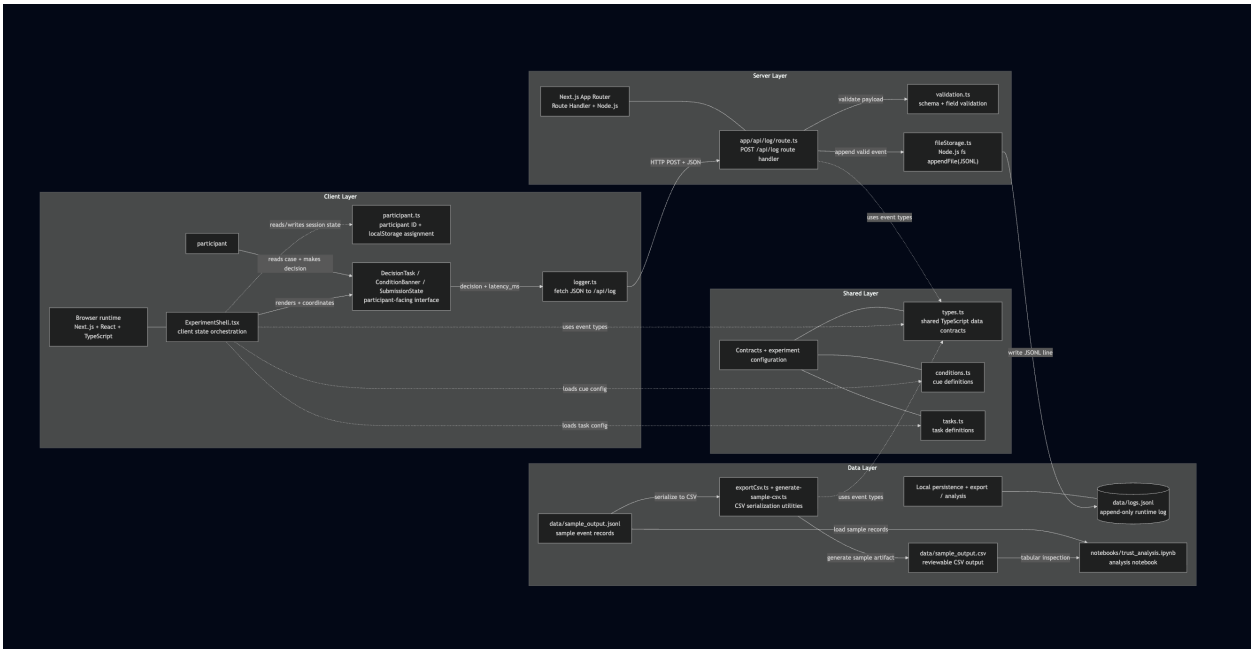

Figure 1: System design for the proposed HumanAI trust experimentation platform.

3.3 Prototype Homepage Mockup

Figure 2 shows a sample homepage for the proposed experiment interface. The homepage demonstrates the intended user-facing structure of the prototype: a lightweight experimental shell, clear presentation of the assistant condition, and a focused interface for the recommendation-acceptance task. This mockup reflects the project goal of building a clean, participant-friendly frontend while keeping the implementation modular and research-oriented.

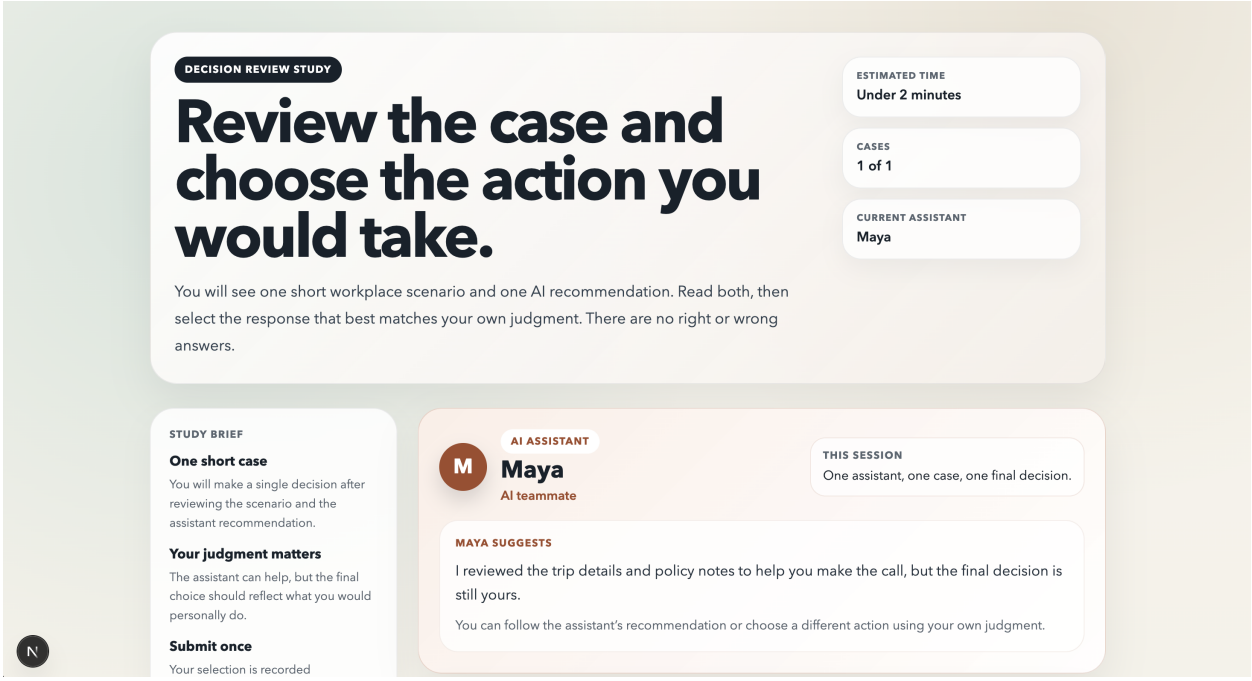

Figure 2: Sample homepage mockup for the experimental web interface.

4 Project Deliverables

4.1 Objective and Implementation Details

The objective of this project is to build a reusable web-based experiment platform for behavioral trust studies in AI-assisted decision making.

Required Deliverables

- Modular web-based experimental prototype
- Participant ID assignment and randomized condition assignment
- Cue manipulation system supporting at least 3 cue dimensions
- One recommendation-acceptance behavioral task
- Structured behavioral logging backend
- Exportable JSON and CSV datasets
- Event schema documentation
- Reproducible Python analysis notebook
- Public GitHub repository with setup and run instructions

Optional Deliverables

- Additional cue dimensions such as visual identity or role framing
- Optional post-task trust/confidence ratings
- A second task type
- Richer interaction logging

4.2 Timeline for Deliverables

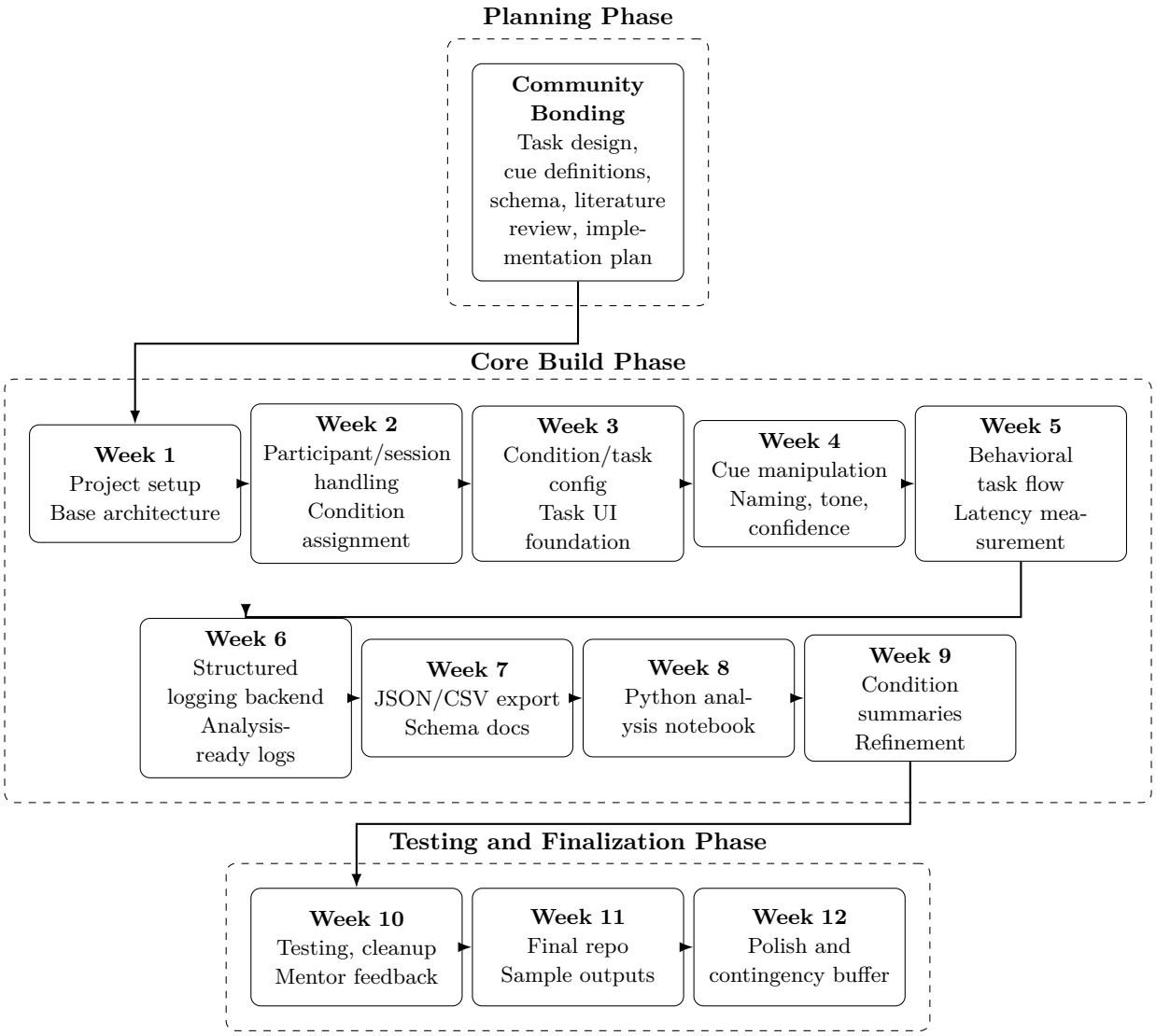

Figure 3: Proposed implementation timeline organized by project phase.

5 Other Information

5.1 Why ISSR / HumanAI

This project strongly aligns with my interests in the psychological aspects of AI assistants, and provides a modality for me to use my previous experience with web development to explore this further.

5.2 Past Experience

I have built a variety of webapps through my work at HTF and Jumpseat. Additionally, I am an incoming SWE intern @ Netflix this summer where I will be working on web-based logging tools, which are all very pertinent to this project.